School Curriculum and Standards Authority

JUDGING STANDARDS IN YEAR 10 MATHEMATICS

Reporting against the Achievement Standard

These assessment pointers are for judging standards of student performance in Year 10 Mathematics. They are examples of what *students may demonstrate* rather than being a checklist of *everything they should do*. For reporting, they are used to make on balance judgments about achievement that will depend on what has been taught and assessed *during the reporting period.* They can also be used for guiding the pitch of assessment tasks, developing marking keys and informing assessment feedback.

YEAR 10 MATHEMATICS ACHIEVEMENT STANDARD

By the end of Year 10, students recognise the connection between simple and compound interest. They solve problems involving linear equations and inequalities. They make the connections between algebraic and graphical representations of relations. Students solve surface area and volume problems relating to composite solids. They recognise the relationships between parallel and perpendicular lines. Students apply deductive reasoning to proofs and numerical exercises involving plane shapes. They compare data sets by referring to the shapes of the various data displays. They describe bivariate data where the independent variable is time. Students describe statistical relationships between two continuous variables. They evaluate statistical reports.

Students expand binomial expressions and factorise monic quadratic expressions. They find unknown values after substitution into formulas. They perform the four operations with simple algebraic fractions. Students solve simple quadratic equations and pairs of simultaneous equations. They use triangle and angle properties to prove congruence and similarity. Students use trigonometry to calculate unknown angles in right-angled triangles. Students list outcomes for multi-step chance experiments and assign probabilities for these experiments. They calculate quartiles and inter-quartile ranges.

YEAR 10 MATHEMATICS ASSESSMENT POINTERS

	A Excellent achievement	B High achievement	C Satisfactory achievement	D Limited achievement	E Very low achievement
Number and Algebra	Extracts relevant information from word problems and applies repeated interest calculations for the comparison of loans and investments. Provides valid justification for the recommendations made. Uses inverse calculations to solve for the rate with compound interest, showing full working.	Extracts multiple pieces of information from text and tables to make ongoing interest calculations for loans and investments. Uses inverse calculations to solve for the principal in compound interest.	Applies formulas for simple and compound interest. Identifies compound interest as repeated applications of simple interest.	Applies simple interest to all calculations.	Attempts to apply simple interest, but makes frequent errors.

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	A Excellent achievement	B High achievement	C Satisfactory achievement	D Limited achievement	E Very low achievement
Number and Algebra	Simplifies sums and differences of algebraic fractions with unrelated denominators. Applies index laws to simplify complex algebraic products and quotients involving integral indices. Substitutes into formulas and solves the resulting quadratic equation in a variety of ways, identifying that some solutions may be inappropriate.	Simplifies sums and differences of algebraic fractions with related denominators. Substitutes into formulas, identifying the form of the resulting equation, and selects the appropriate method of solution. Applies index laws to simplify algebraic products and quotients involving integral indices.	Simplifies sums and differences of algebraic fractions with like denominators. Substitutes into formulas and solves the resulting linear equation. Applies index laws to simplify algebraic products and quotients involving non-negative integral indices.	Substitutes values into formulas to determine an unknown. Applies index laws to simplify numerical expressions with positive integral powers, but makes frequent errors when applying index laws to algebraic expressions.	Substitutes values into simple formulas. Shows the meaning of a number written with a power, and evaluates it.
	Expands and simplifies the sums and differences of binomial expressions involving fractions. Applies factorisation multiple times to expressions including non-monic quadratic trinomials. Applies multi-step factorisation, at least three times, to simplify products and quotients of algebraic fractions. Formulates a quadratic equation by extracting relevant information from a complex word problem. Rearranges the equation into a form for ease of computation and solves it through a variety of techniques, including graphical. Applies the quadratic formula, showing full working, to obtain the simplified irrational solution/s.	Expands and simplifies the sums and differences of binomial expressions. Factorises two-step algebraic expressions involving common factors, quadratic binomials and monic trinomials. Applies one-step factorisation to simplify algebraic products and quotients. Solves a quadratic equation algebraically and justifies the solution/s using a variety of techniques.	Expands and simplifies, where necessary, binomial expressions. Factorises monic quadratic expressions such as common factors, binomials, trinomials and quadnomials. Solves a simple quadratic equation algebraically and links the solution to the x- intercepts on the matching graph. Applies the four operations to simple algebraic fractions. Simplifies algebraic fractions expressed in factored form.	Solves a quadratic equation graphically and when given in factored form. Factorises a one-step algebraic expression involving a numerical common factor.	Solves linear equations with up to three steps only.

	A Excellent achievement	B High achievement	C Satisfactory achievement	D Limited achievement	E Very low achievement
Number and Algebra	Formulates a pair of simultaneous linear equations from a word problem and determines the simultaneous solution in a variety of ways. Rearranges linear rules given in a variety of forms to justify the presence, or not, of a geometric relationship. Solves problems involving perpendicular lines. Formulates and solves a linear equation or inequality involving fractions from a complex word problem, verifies the solution/s and illustrates the solutions of inequalities with a suitable graph.	Graphs a pair of linear relations given in a variety of ways, writes the simultaneous solution and justifies algebraically. Rearranges linear rules into forms to make conclusions about the geometric relationship. Solves problems involving parallel lines. Uses algebraic techniques to solve a pair of simultaneous linear equations. Solves problems involving linear equations and inequalities involving fractions. Illustrates the solution of inequalities graphically.	Graphs a pair of simple linear relations and writes the simultaneous solution. Identifies the relationship between lines as being parallel or perpendicular by referring to the gradients. Solves problems involving linear equations or inequalities.	Writes the simultaneous solution when given a graph of a pair of linear relations.	Solves linear equations involving one variable.
	Makes connections between transformed relations and writes the rule for the given graph.	Makes connections between the algebraic and graphical forms of quadratic, circular and exponential relations, distinguishing between rules for relations of the same kind. Sketches relations, applying transformations.	Makes connections between the algebraic and graphical representations of relations such as quadratic, circular and exponential.	Makes connections between the algebraic and graphical representation of linear and quadratic relations.	Makes connections between the algebraic and graphical representation of linear relations only.
Measurement and Geometry	Draws a diagram to represent information extracted from a multi-step word problem. Solves problems by applying the appropriate surface area and volume formulas for objects, including inverse use of formulas and rates.	Draws a diagram to represent information extracted from a word problem. Solves problems involving composite solids by applying the appropriate surface area and volume formulas.	Uses the appropriate formulas to calculate the surface area and volume of objects comprising of prisms.	Calculates the surface area and volume of a simple right prism.	Attempts to calculate the surface area and volume of a simple right prism, but confuses concepts and makes consistent errors.

	A Excellent achievement	B High achievement	C Satisfactory achievement	D Limited achievement	E Very low achievement
Measurement and Geometry	Justifies similarity of triangles after drawing a diagram to represent a familiar two-dimensional situation, such as trees and shadows, and applies ratios to deduce the unknown. Draws a diagram for given information, proves congruency or similarity and deduces further properties of geometric figures.	Draws a diagram to represent a familiar two-dimensional situation, such as trees and shadows, and applies ratios to deduce the unknown. Draws a diagram for given information and proves congruency or similarity of triangles.	Uses angle and triangle properties to prove congruency or similarity of triangles using given diagrams. Applies ratios and geometric reasoning to deduce missing values in given diagrams.	Uses ratios to deduce unknowns when given a pair of similar triangles. Applies congruency to triangles to determine unknown values.	Identifies pairs of similar triangles. Identifies pairs of congruent triangles.
	Chooses to use algebra in solving a problem requiring the use of Pythagoras' Theorem to find the missing lengths of the side/s. Diagrammatically represents a complex word problem, including the application of direction and elevation. Follows multiple steps to solve the problem, e.g. applies rates, Pythagoras' Theorem and trigonometry.	Diagrammatically represents a word problem involving elevations. Follows multiple steps to solve the problem, e.g. applies Pythagoras' Theorem and trigonometry.	Selects and uses the correct trigonometric ratio to calculate unknown angles.	Uses trigonometric ratios to find unknown sides in right-angled triangles.	Identifies some trigonometric ratios.
Statistics and Probability	Lists outcomes for multi-step chance experiments, with and without replacement, including the use of probability trees, assigns probabilities to the outcomes and determines probabilities of compound events, with conditions. Identifies events that are independent.	Lists outcomes for multi-step chance experiments, with and without replacement, assigns probabilities to the outcomes and determines probabilities of compound events. Draws two-way tables and Venn diagrams, and determines probabilities of compound events.	Lists outcomes for multi-step chance experiments and assigns probabilities.	Lists outcomes for two-step chance experiments and determines associated probabilities.	With assistance, lists outcomes for one- and two-step chance experiments, and determines associated probabilities.

	A Excellent achievement	B High achievement	C Satisfactory achievement	D Limited achievement	E Very low achievement
Statistics and Probability	Matches box plots to corresponding histograms and dot plots, and provides an explanation using concise statistical reasoning. Applies algebraic techniques to calculate the equation of the trend line for bivariate and time-series data. Uses the equation to make predictions, identifying factors that influence the reliability of predictions and gives reasons why predictions and actual values may vary. Uses explicit statistical language when writing statistical reports, making links/references to resources relevant to the report.	Uses the quartiles, together with minimum, median and maximum data values, to draw box plots of data from primary and secondary sources, identifying possible outliers. Compares data presented on parallel box plots, using correct statistical language. Investigates and describes relationships between variables for bivariate and time-series data relating to economic, social and environmental issues. Draws trend lines, comments on predictions and gives possible reasons why predictions and actual values may vary.	Calculates quartiles and interquartile range for collected or given authentic data. Compares data sets by referring to the shape of the display. Describes the trend and patterns for bivariate data where the independent variable is time. Describes the statistical relationship between two continuous variables. Evaluates statistical reports.	Calculates quartiles and interquartile range for a small sample of ordered data. Completes or draws a scatter plot for bivariate or time-series data. Writes statistical reports using vague language.	Completes a scatter plot and calculates the mean and median for each variable.